

Succession

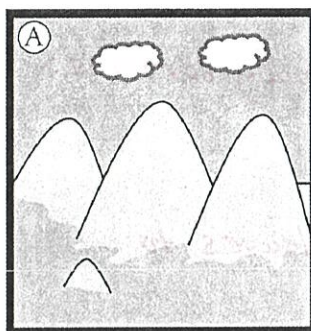
How do ecosystems develop over time?

Key

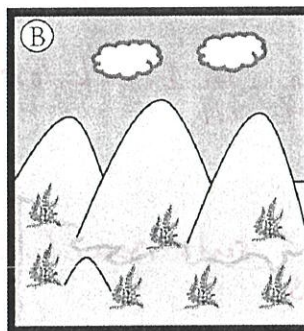
Why?

On May 18, 1980, Mount St. Helens in the state of Washington erupted with the force of a hydrogen bomb. The volcano had been dormant for over 120 years, but now 57 people were dead and forests and lakes were totally destroyed, including nearby Spirit Lake, which became a mud hole. The blast leveled trees in areas over 10 miles from the crater and ash deposits suffocated life on the mountain. However, within weeks, mammals that had taken shelter underground started to reappear in the area, and now, over thirty years later, many areas of the mountain are colonized with a large variety of plant and animal life. How does an area move from a sterile, barren wilderness to one full of life?

Model 1 – Primary Succession

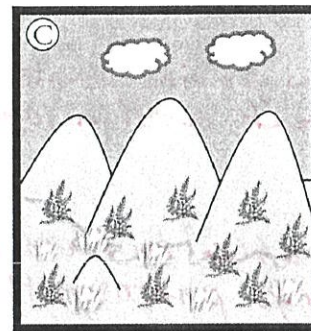


Barren rock from beneath a retreating glacier, or due to a volcanic eruption.



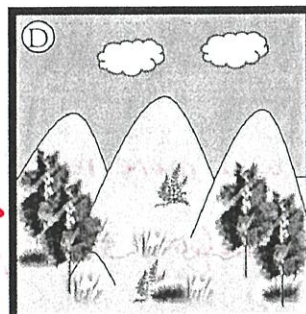
Low-growing plants such as mosses, ferns, and lichens begin to colonize.

Pioneer community

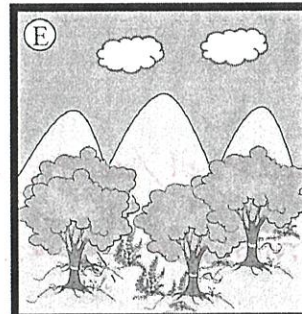


Fast-growing grasses, flowering plants, and small shrubs begin to take root. A thin layer of soil develops.

Seral stages →



Fast-growing trees such as birch and mountain ash form a low forest and shade out lower plants.



100–200 years since colonization began, large, slow-growing trees, such as an oak, become established.

Climax community

1. Refer to Model 1.

- a. On what type of land does primary succession first begin to occur?

Barren rock beneath a glacier melt or volcanic eruption

- b. Does there appear to be any life on the land when primary succession begins?

None

- c. Why would most plants such as shrubs and trees find it difficult to grow here?

There is not enough soil (& nutrients) for its needs

2. Refer to diagram B in Model 1.

- a. What are the first organisms (colonizers) on this land?

Low growing plants like mosses, ferns, & lichens

- b. Suggest the mechanisms by which the first colonizers arrived on the land.

Wind brings in spores/seeds

Read This!

The first colonizers are referred to as the **pioneer community**. These can include lichens, mosses, ferns, and bacteria—all organisms with low nutrient requirements. As they colonize, they break the weathered rock surface, which helps to create the first thin layer of soil. Without soil other plant life cannot be sustained and without plants no animal life can exist.

3. Refer to Model 1.

- a. Which diagram illustrates a pioneer community? *Diagram B*

- b. What are some of the features of the pioneer community?

Low growing plants that can breakdown rocky surfaces to create a thin layer of soil.

4. Notice the colonizers in diagrams C and D are taller and require more nutrients than those in the pioneer community. Considering what you already know about plants and photosynthesis, why might it be a competitive advantage for a plant to be taller?

To be able to have more sun reach its leaves to undergo photosynthesis.

5. What happens to the pioneer organisms once the new colonizers become established?

They are outcompeted for resources and probably do not live. (adds to soil)

Read This!

As the newer colonizers begin to take over, animals will also begin to appear so they can feed on the more diverse food source. The pioneer plants die and decompose and the animals leave behind manure. Both add to the thin soil layer.

6. What effect will the addition of animal waste and decayed plant matter have on the soil and land?

Increase the soil level/depth and add more nutrients to it.

7. How will grazing animals help plants to become established?

by eating grasses that may not last long helping larger plants obtain resources

8. How will the grazing animals prevent or control further colonization by other plants?

Will eat them as they try to take root/grow.

9. Using the diagrams in Model 1 as a guide to develop a definition with your group for the term primary succession, as it relates to the colonizing of barren land.

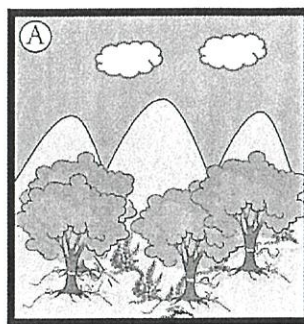
Primary succession begins where no soil has ever existed. pioneer species have to settle & begin to add soil from rock. As more soil is laid down then larger & larger plants begin to grow.

Read This!

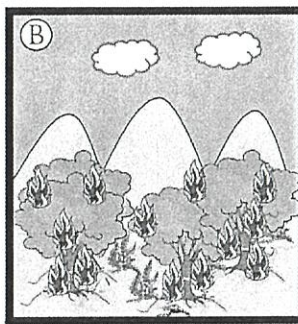
As soil quality and quantity improves, the life forms present in the area undergo a series of changes, each referred to as a **seral stage**. Eventually a stable climax community is formed.

10. Label the pictures in Model 1 as pioneer community, seral stages, and climax community.

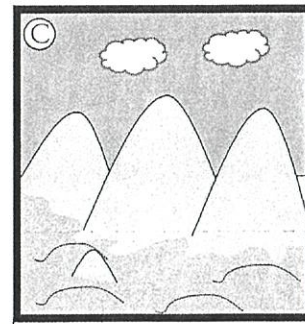
Model 2 – Secondary Succession



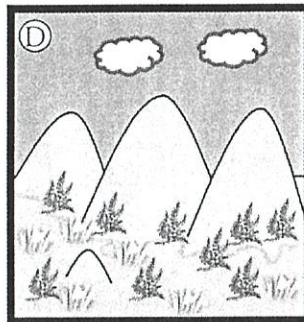
Established climax community.



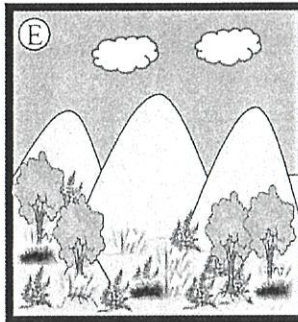
Forest fire.



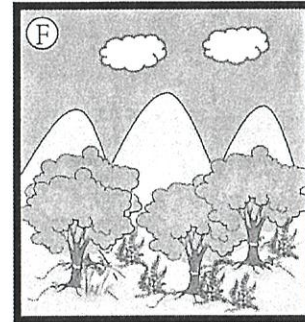
Plants have been destroyed and animals have fled the burned out area. A layer of ash is over the soil.



After 3–5 years grasses and low-growing shrubs have colonized the land.



After 20 years small trees form young woodland.



50–100 years after the setback event, mature oak woodland is restored.

11. Refer to the diagrams in Model 2.

- What stage of development does diagram A represent? *Climax community.*
- What appears to have happened in diagram B?
A forest fire
- What could be two causes of this event?
Lightning may struck a tree. Humans being careless
- What process will begin again after this event has occurred?
Colonization of the land w/ pioneer species.

12. Can the ecosystem totally recover from this set-back? What evidence is given in Model 2?

Yes, because a layer of ash is over the soil in diagram C this allows succession to start with grasses & shrubs

13. Why is the title of Model 2 Secondary Succession rather than Primary Succession?

Because at one point plants had already been established on land with soil.

14. Consider each event below and determine if the recovery process for the environment will happen by primary succession or secondary succession.

- Melting, receding glaciers = *PRIMARY*
- Logging a wooded area = *Secondary*
- Major flooding of a creek bed = *Secondary*
- Volcanic eruption with lava flow = *PRIMARY*

1. The first part of the paper is devoted to a discussion of the

main results of the paper.

2. The second part of the paper is devoted to a discussion of the

main results of the paper.

3. The third part of the paper is devoted to a discussion of the

main results of the paper.

4. The fourth part of the paper is devoted to a discussion of the

main results of the paper.

5. The fifth part of the paper is devoted to a discussion of the